

Basic Rules of Probability

① Important Definitions

In probability, understanding the basic definitions is crucial for grasping more complex concepts.

- **Experiment:** An action or process that leads to one or more outcomes (e.g., rolling a die). Deterministic experiments have predictable outcomes, while probabilistic or random experiments have uncertain outcomes.
- **Outcome:** A possible result of an experiment (e.g., rolling a 3 on a die).
- **Probability (P):** A measure of the likelihood that an event will occur, ranging from 0 (impossible) to 1 (certain).
- **$P(A \cup B)$:** The probability of event A or event B occurring.
- **$P(A \cap B)$:** The probability of both events A and B occurring.
- **Sample Space (S):** The set of all possible outcomes of an experiment.
- **Event (E):** A subset of the sample space.
- **Mutually Exclusive Events:** Two events that cannot occur at the same time (e.g., rolling a 2 and rolling a 5 on a die).

$$P(A \cap B) = 0$$

- **Independent Events:** Two events where the occurrence of one does not affect the other.

$$P(A \cap B) = P(A) \cdot P(B)$$

- **Certain Event:** An event that is guaranteed to happen, with a probability of 1.
- **Impossible Event:** An event that cannot happen, with a probability of 0.
- **Exhaustive Events:** A set of events that cover the entire sample space, meaning at least one of them must occur.

Example:

- Tossing a die:
 - Sample space: $S = \{1, 2, 3, 4, 5, 6\}$
 - Event (E): Rolling an even number = $\{2, 4, 6\}$

2. The Complement Rule

- The complement of an event A is the event that A does not occur.
- Notation: A^c
- Rule: $P(A^c) = 1 - P(A)$
- Example: If the probability of rain today is 0.3, the probability it won't rain is:
 - $P(\text{No Rain}) = 1 - 0.3 = 0.7$

3. The Addition Rule

- Used to calculate the probability of the union of two events.
- For general events:
$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$
- For mutually exclusive events:
$$P(A \text{ or } B) = P(A) + P(B)$$
- Example:

$$P(A) = 0.4, \quad P(B) = 0.5, \quad P(A \text{ and } B) = 0.2$$

$$P(A \text{ or } B) = 0.4 + 0.5 - 0.2 = 0.7$$

④ The Multiplication Rule

Used to find the probability that two events occur together.

- For Independent events:

$$P(A \text{ and } B) = P(A) \times P(B)$$

Example: A event A is rolling a 3 on a first throw of a die, and event B is rolling an even number on second throw.

- $P(A) = \frac{1}{6}$ (rolling a 3)
- $P(B) = \frac{3}{6}$ (rolling a 2, 4, or 6)
- $P(A \text{ and } B) = P(A) \times P(B) = \left(\frac{1}{6}\right) \times \left(\frac{3}{6}\right) = \frac{1}{12}$

- For Dependent events:

$$P(A \text{ and } B) = P(A) \times P(B|A)$$

Example: A bag contains 5 red and 3 blue balls. Two balls are drawn without replacement. Let:

- Event A be drawing a red ball first.
- Event B be drawing a red ball second.
- $P(A) = \frac{5}{8}$ (5 red balls out of 8)

After removing one red ball, there are 4 red left out of 7 total:

$$P(B|A) = \frac{4}{7}$$

- $P(A \text{ and } B) = \left(\frac{5}{8}\right) \times \left(\frac{4}{7}\right) = \frac{20}{56} = \frac{5}{14}$

So the probability of drawing two red balls without replacement is $\frac{5}{14}$.

5. Independent vs Dependent Events - More examples

- Independent: The outcome of one event does not affect the other.
→ Example: Tossing two coins.
- Dependent: One event affects the probability of the other.
→ Example: Drawing two cards without replacement.

6. Mutually Exclusive Events

Two events are mutually exclusive if they cannot happen at the same time.

Example:

- Event A: Rolling a 2
- Event B: Rolling a 5

These are mutually exclusive because a die can't show both at once.

Summary

This lesson covered key probability rules and concepts that are foundational for more advanced topics like distributions and statistical inference.

- ★ Complement Rule: $P(A^c) = 1 - P(A)$
- ★ Addition Rule for unions
- ★ Multiplication Rule for intersections
- ★ Understanding independence and exclusivity

Homework / Practice

- ① A card is drawn from a standard deck. What is the probability of drawing a red card or a queen?
- ② If two dice are rolled, what is the probability that both show even numbers?
- ③ Think of an example from your daily life where two events are dependent.

≡ Practice Questions ≡

- ① A card is drawn from a standard deck. What is the probability of drawing a red card or a queen?

Solution:

Let A = red card, B = queen.

$$P(A) = \frac{26}{52} = \frac{1}{2} \quad (26 \text{ red cards})$$

$$P(B) = \frac{4}{52} = \frac{1}{13} \quad (4 \text{ queens})$$

$$P(A \cap B) = \frac{2}{52} = \frac{1}{26} \quad (\text{red queens: } Q\heartsuit, Q\spadesuit)$$

Using inclusion-exclusion:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{13} - \frac{1}{26}$$

$$= \frac{13}{26} + \frac{2}{26} - \frac{1}{26} = \frac{14}{26} = \frac{7}{13}$$

$$\therefore P(\text{red card or queen}) = \frac{7}{13}$$

- ② If two dice are rolled, what is the probability that both show even numbers?

Solution:

Each die has 3 even numbers: 2, 4, 6 $\rightarrow P(\text{even}) = \frac{3}{6} = \frac{1}{2}$

Since the dice are independent,

$$P(\text{both even}) = P(\text{even on first}) \times P(\text{even on second})$$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$\therefore P(\text{both even}) = \frac{1}{4}$$

- ③ Think of an example from your daily life where two events are dependent.

Solution:

Example: If you draw a card from a deck, the event "the card is a heart" and the event "the card is a face card" are dependent.

Reason: If the card is a heart, the probability that it is a face card is higher (3 out of 13 hearts are face cards), whereas if the card is not a heart, the probability is lower (9 out of 39).

$\rightarrow$ So, the two events are dependent.

The solutions:

- ① Box with 4 red, 3 green, 2 blue balls — $P(\text{not green})$?

$$\text{Total balls} = 4 + 3 + 2 = 9$$

$$P(\text{green}) = \frac{3}{9} = \frac{1}{3}$$

$$P(\text{not green}) = 1 - \frac{1}{3} = \boxed{\frac{2}{3}}$$

- ② Two coins tossed — $P(\text{at least one head})$?

Sample space: $\{HH, HT, TH, TT\} \rightarrow 4$ outcomes

$$P(\text{no heads}) = P(TT) = \frac{1}{4}$$

$$P(\text{at least one head}) = 1 - \frac{1}{4} = \boxed{\frac{3}{4}}$$

- ③ Number from $\{1, \dots, 10\}$ — $P(\text{multiple of 3 or 5})$?

Multiples of 3: $\{3, 6, 9\} \rightarrow 3$ numbers

Multiples of 5: $\{5, 10\} \rightarrow 2$ numbers

No overlap (nothing is a multiple of both 3 and 5 in this set)

$$\text{Favorable} = 3 + 2 = 5$$

$$P = \frac{5}{10} = \boxed{\frac{1}{2}}$$

- ④ Bag with 6 white, 4 black balls — draw two with replacement — $P(\text{both white})$?

Total = 10 balls

Since it's with replacement, each draw is independent:

$$P(\text{white}) = \frac{6}{10} = \frac{3}{5}$$

$$P(\text{both white}) = \frac{3}{5} \times \frac{3}{5} = \boxed{\frac{9}{25}}$$

- ⑤ Student knows 70% of answers — $P(\text{doesn't know the answer})$?

$$P(\text{knows}) = 0.70$$

$$P(\text{doesn't know}) = 1 - 0.70 = \boxed{0.30 \text{ (or } \frac{3}{10})}$$